ECEN 662 Estimation & Detection Theory **Name:**
Spring 2026 — Homework #7
Due date: Sunday, April 12

Problem 1 (Bayes estimation, 20 Points): Let $X = (X_1, \ldots, X_n)^t$ where

$$X_i \overset{iid}{\sim} Bernoulli(\theta),$$

where θ is the unknown parameter. Assume that the prior distribution for Θ is $\mathcal{U}((0,1))$.

i) (10 Points) Find the Bayes estimator of θ under the squared-error loss.

ii) (10 Points) Now let X_{n+1} be a new observation such that $X_{n+1} \sim Bernoulli(\theta)$ and is independent of X given θ. Find $\mathbb{P}(X_{n+1} = 1|X)$.

Problem 2 (Stein's lemma, 20 Points): Suppose that X has a density of exponential form

$$f_\theta(x) \propto_x h(x) \exp\left(\sum_{i=1}^{d} \theta_i T_i(x) - A(\theta)\right)$$

where each $T_i(x)$ is differentiable.

i) (10 Points) Show that if X has support in real line and g is any differentiable function such that $\mathbb{E}_\theta[g'(X)] < \infty$, then

$$\mathbb{E}_\theta\left[\left(\frac{h'(X)}{h(X)} + \sum_{i=1}^{d} \theta_i T_i'(X)\right) g(X)\right] = -\mathbb{E}_\theta[g'(X)].$$

ii) (5 Points) Using part a), show that if $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\mathrm{Cov}_\theta[g(X), X] = \sigma^2 \mathbb{E}_\theta[g'(X)]$$

where $\theta = [\mu, \sigma^2]^t$.

iii) (5 Points) Use part ii) to compute the third and fourth moments of $\mathcal{N}(\mu, \sigma^2)$.

Problem 3 (Tweedie's formula, 20 Points): Let

$$f_\theta(x) = h(x) \exp(\theta x - A(\theta))$$

be a one-parameter, canonical exponential family.

i) (10 Points) Show that for any prior $\pi(\theta)$,

$$\mathbb{E}_\pi[\Theta|X = x] = \frac{d}{dx} \log\left(\frac{m_\pi(x)}{h(x)}\right),$$

where $m_\pi(x)$ is the marginal distribution of X.

ii) (10 Points) Use part i) to prove Tweedie's formula:

$$\mathbb{E}_\pi[\Theta | X = x] = x + \frac{d}{dx} \log m_\pi(x),$$

when f_θ is $\mathcal{N}(\theta, 1)$.

Problem 4 (Empirical Bayes estimation, 20 Points): Assume that

$$\Theta_i \overset{iid}{\sim} \pi(\theta_i) \quad \text{and} \quad X_i | \Theta_i = \theta_i \overset{ind}{\sim} Poisson(\theta_i).$$

Show that for any prior π and any $i \in [n]$, we have

$$\mathbb{E}_\pi[\Theta_i | X_i = x_i] = \frac{(x_i + 1) m_\pi(x_i + 1)}{m_\pi(x_i)} = \frac{(x_i + 1)\mathbb{E}_\pi[Y_{x_i+1}]}{\mathbb{E}_\pi[Y_{x_i}]},$$

where $Y_x = \sum_{i=1}^n 1_{\{X_i = x\}}$.